

Project Development Phase

No. of Functional Features Included in the Solution

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

The table below lists every functional feature implemented in the working solution, mapped to the file(s) where it is implemented.

#	Functional Feature	Description	Implemented In
1	Historical dataset loading & cleaning	Imputing missing values, removing duplicates, and correcting data types	Notebooks/EDA_and_Model_Training.ipynb
2	Exploratory Data Analysis	Examines class balance, feature distributions, correlation heatmap, and outliers (8 charts produced)	Notebooks/EDA_and_Model_Training.ipynb, Assets/
3	Feature encoding & scaling	Categorical fields label-encoded and numeric fields scaled with StandardScaler	Notebooks/EDA_and_Model_Training.ipynb
4	Multi-model training	Trains and compares Logistic Regression, Decision Tree, Random Forest, and XGBoost	Notebooks/EDA_and_Model_Training.ipynb
5	Model evaluation	Evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Classification Report, and ROC Curve	Notebooks/EDA_and_Model_Training.ipynb
6	Best-model selection	An automated comparison table picks XGBoost as the best model based on F1-Score	Notebooks/EDA_and_Model_Training.ipynb
7	Model persistence	The trained model, encoders, scaler, and feature ordering are saved with joblib	Model/*.pkl
8	Web-based application form	A responsive HTML/Bootstrap form used to enter applicant details	Flask/templates/index.html
9	Live prediction endpoint	The POST /predict route executes the saved model and returns an Approved/Rejected result	Flask/app.py
10	Confidence score display	Displays the model's approval probability next to the prediction result	Flask/app.py, Flask/templates/result.html

#	Functional Feature	Description	Implemented In
11	Input validation	Checks required fields and invalid categories, showing clear error messages to the user	Flask/app.py, Flask/templates/index.html
12	Custom error handling	A custom 404 page replaces Flask's default error screen	Flask/templates/404.html
13	Logging	Logs model loading status, predictions, and errors using Python's logging module	Flask/app.py
14	Consistent visual branding	A shared navigation bar and footer via Jinja2 template inheritance, with a banking-themed design	Flask/templates/base.html, Flask/static/style.css

Total functional features implemented: 14, spanning data preprocessing, model training and evaluation, and the full end-to-end web application.